

# EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

## Agents hand each other evidence references, not paraphrases.

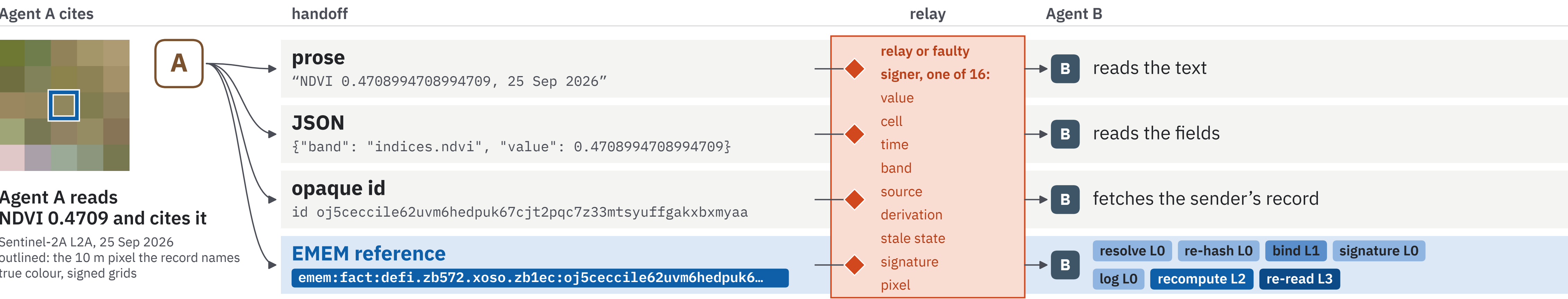
The receiver resolves the reference, re-hashes the record, re-reads the source. A signature fixes the record's bytes, not the measurement.

Jaya Kumari · Avijeeet Singh · Vortex AI · avijeeet@vortex.ai  
Agentic AI for Earth Observation · ESA Φ-lab and BIFOLD · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

1 What survives an agent handoff?

### Handed prose, B acted on 15 of 15 corruptions; handed a reference it checks, on 0 of 16.

A relay or a faulty signer corrupts one signed NDVI record in one of 16 ways; Agent B resolves, re-hashes, binds, checks, recomputes and re-reads.

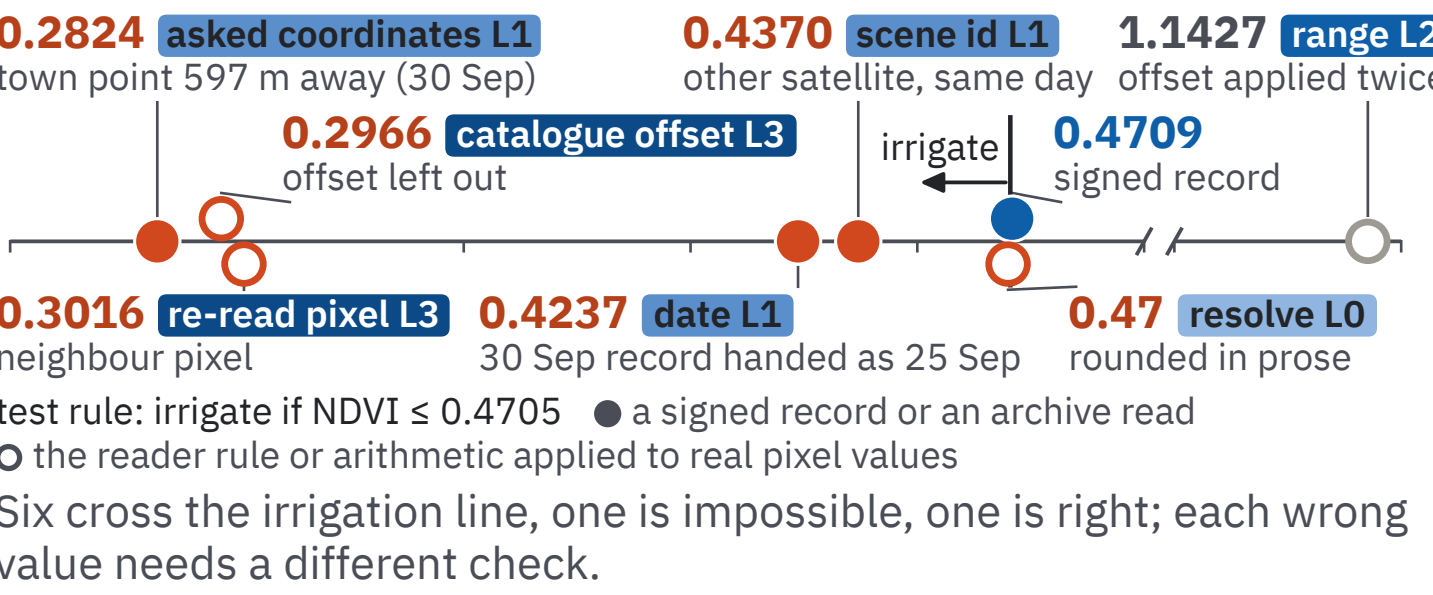


With a model as B and a forged cell, Claude Haiku 4.5 declined 5 of 5; Qwen2.5-3B dropped the cell from all 24 tokens and acted 5 of 5: a reference protects only a receiver that keeps it whole and obeys a refusal.

2 What can an NDVI become between agents?

### Eight answers; one is right

One question, NDVI at the Keylong field on 25 Sep 2026; real records, archive pixels and handling rules give eight values.



3 Why does an agent handoff need a record?

### Our own errors name the checks

Each error happened inside emem, a system built for evidence, or in our tests; each class is documented elsewhere.



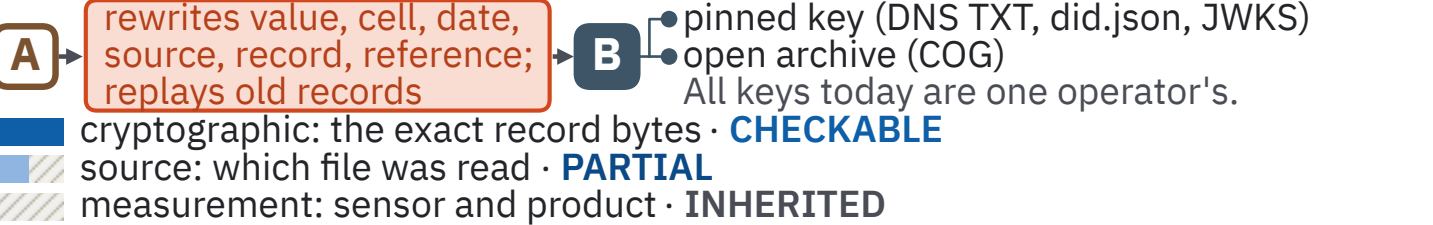
Signatures prevented none; the record made each checkable at a named layer, except the top row.

### We ask four questions

- RQ1 Can a receiver detect corruption without trusting the sender? → 1, 5
- RQ2 Which corruption classes can it detect? → 5
- RQ3 Does a reference beat prose, JSON or an opaque id? → 1
- RQ4 What stays unverifiable with the reference intact? → 6, 7
- H1 A checked reference refuses word-preserving corruptions. H2 Only a source re-read catches a signer's wrong pixel. H3 A historical reference keeps the cited state.

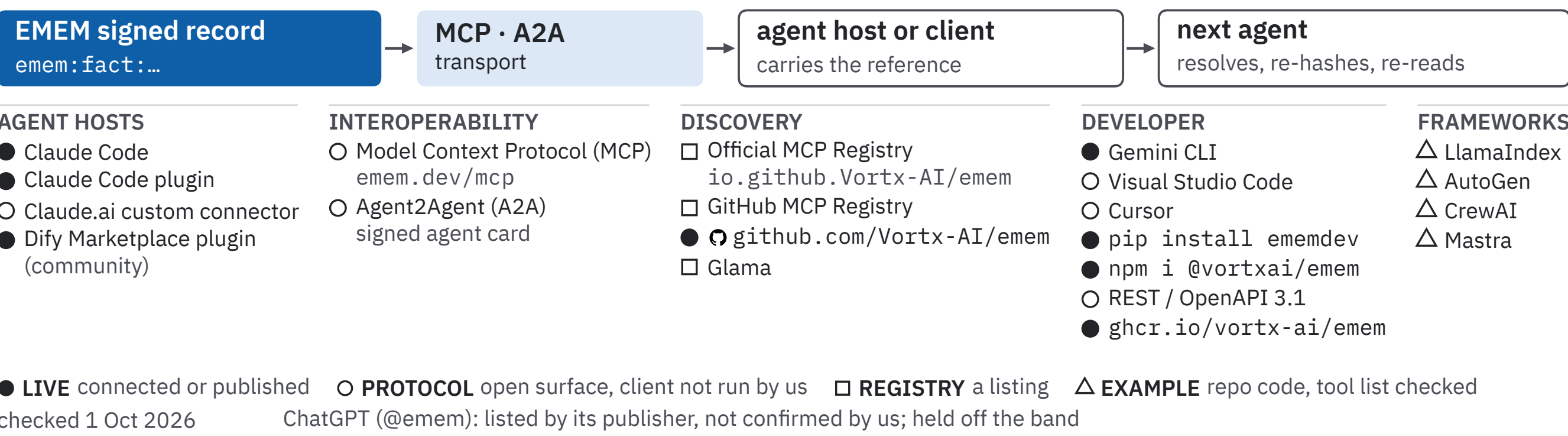
### A relay can rewrite, not sign

A relay can rewrite what it carries but cannot sign under the pinned key or forge an address; a signer's misread pixel is caught only by a re-read. Out of scope: a compromised key, a wrong sensor, the entity meant, the decision.

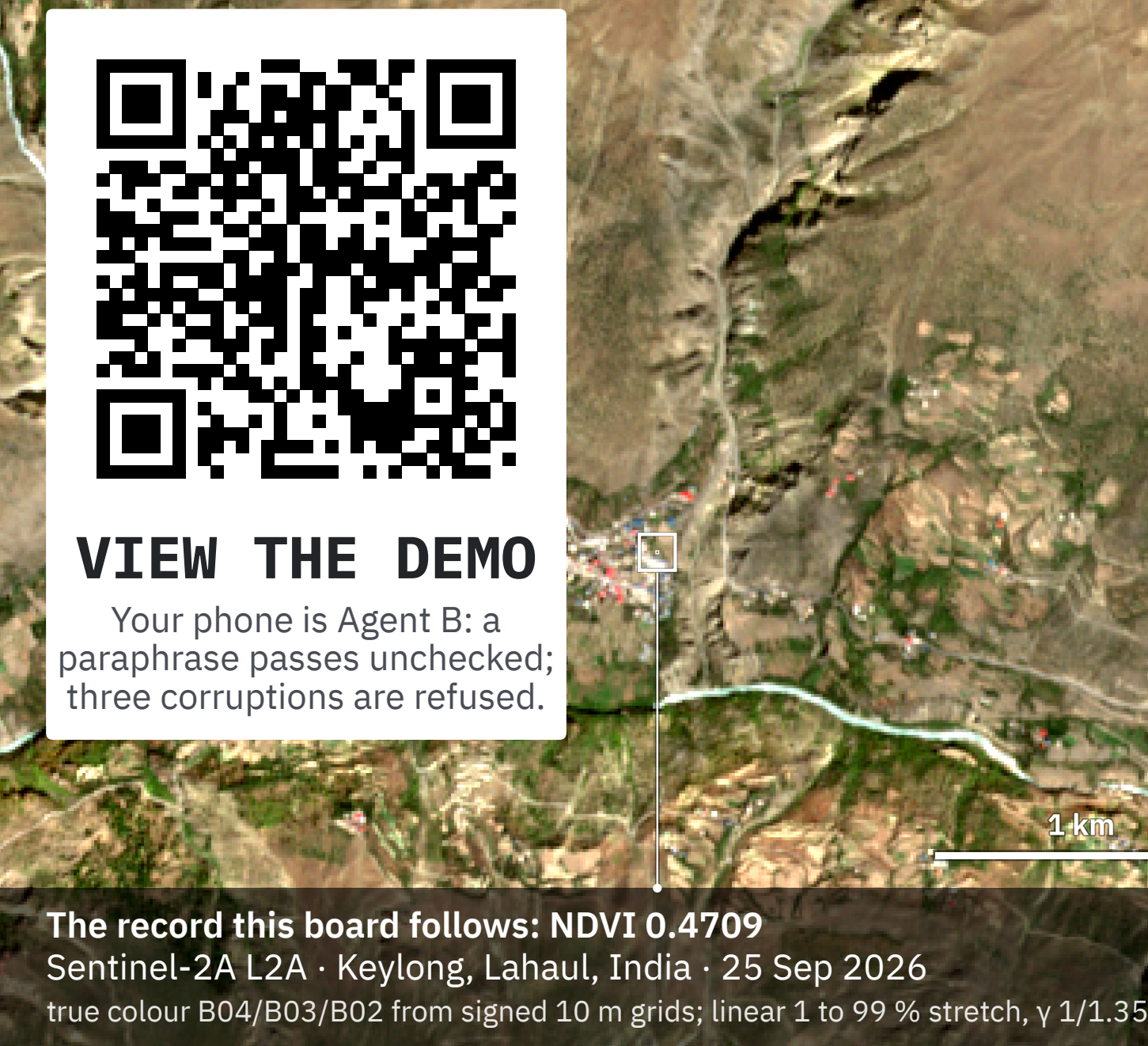


## 12 One evidence protocol, multiple agent runtimes.

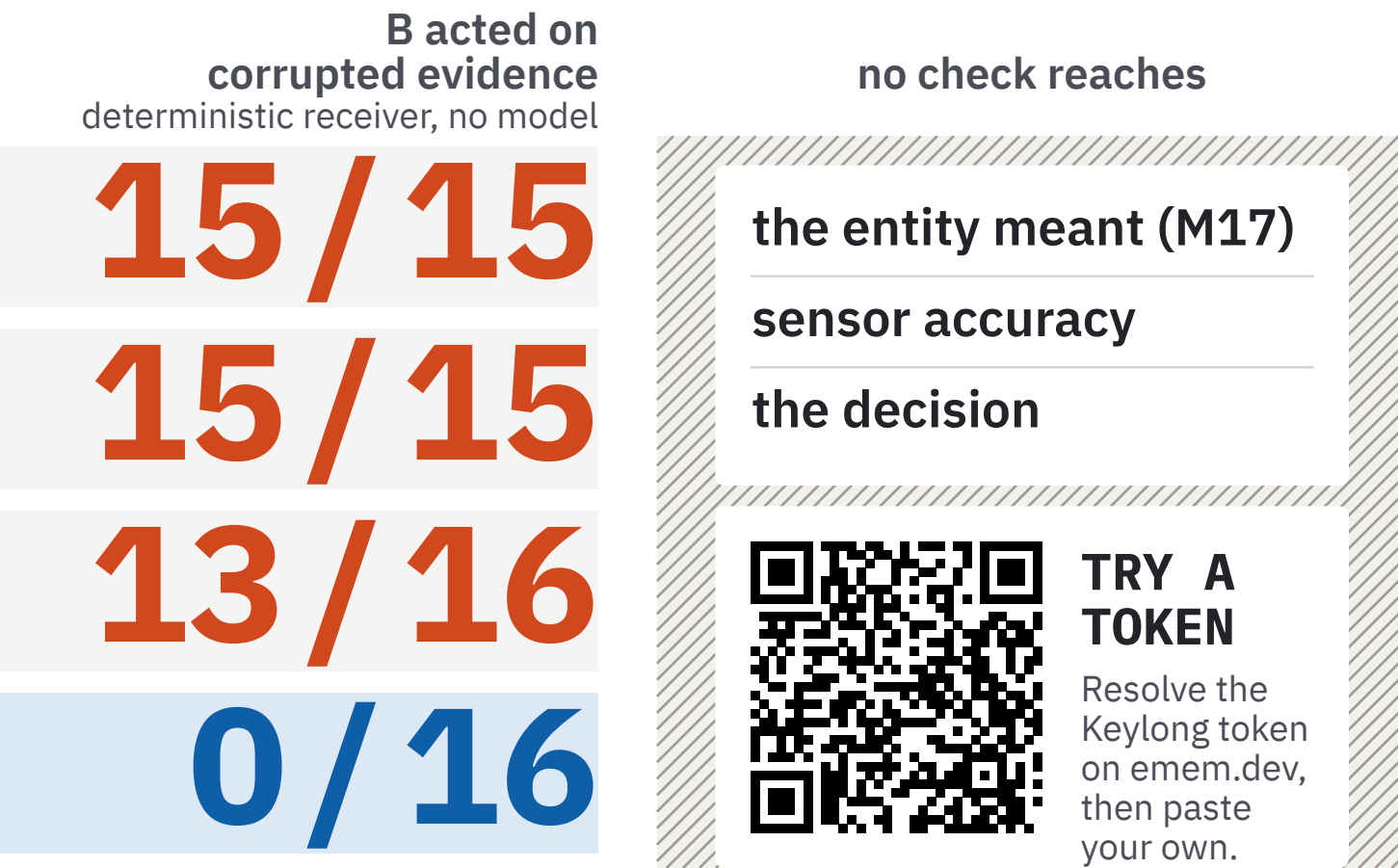
One reference resolved to one address and one value through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026). A listing is not an integration. Not run by us: ChatGPT, claude.ai, Dify, VS Code, Cursor.



Mechanisms read from emem's code; where its docs differ, the code is printed. emem.dev at commit 8e9b401; measurements 29 Sep to 1 Oct 2026. Extending verification to execution: reference harness, no spacecraft enrolled. · Same pattern in software supply chains: Sigstore, RFC 9162, SCITT (RFC 9943). Closest agent memory: ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · Reg. (EU) 2023/1115 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2026-07-28 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Dang et al. 2026, ARC (2607.25066) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)



The record this board follows: NDVI 0.4709  
Sentinel-2A L2A · Keylong, Lahaul, India · 25 Sep 2026  
true colour B04/B03/B02 from signed 10 m grids; linear 1 to 99 % stretch, y 1/1.35



7 What does a checked reference establish?

### Checks stop at the source

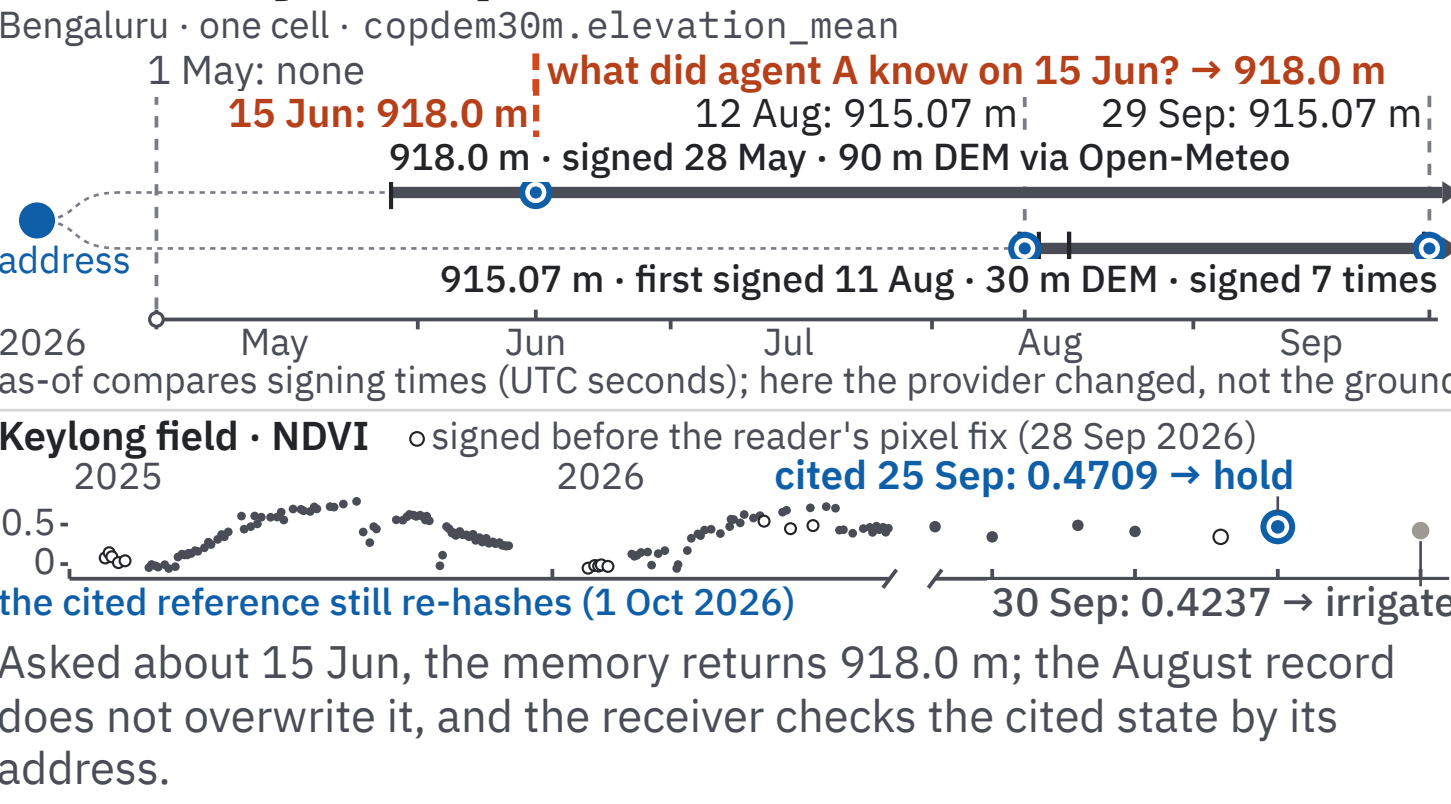
Never "verified" without its layer.

|    |                                                   |                                                                                                                                            |
|----|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| L5 | physical truth, decision INHERITED / OUT OF SCOPE | GFC2020 V3 forest commission error 13.1 %                                                                                                  |
| L4 | entity OUT OF SCOPE                               | M17 passes every check; "this image" became a hair salon                                                                                   |
| L3 | source PARTIAL                                    | named, not hashed: 0 of 213 Keylong sources carry a hash; open archives can be re-read still trusted: that the archive file is the product |
| L2 | derivation RECOMPUTABLE                           | 266 of 780 recomputed bit for signer still trusted: the convention the signature chose                                                     |
| L1 | identity CHECKABLE                                | cell and band bound in 780 of 780; a relabelled token returns 409 still trusted: your own question                                         |
| L0 | record bytes CHECKABLE                            | 780 of 780: hash, signature, log still trusted: the key is emem.dev's                                                                      |

Of 780 records pulled on 30 Sep 2026, 780 pass L0 and L1 and 266 recompute at L2; their source files are named, not hashed. No reference establishes the entity, the sensor's accuracy or the decision.

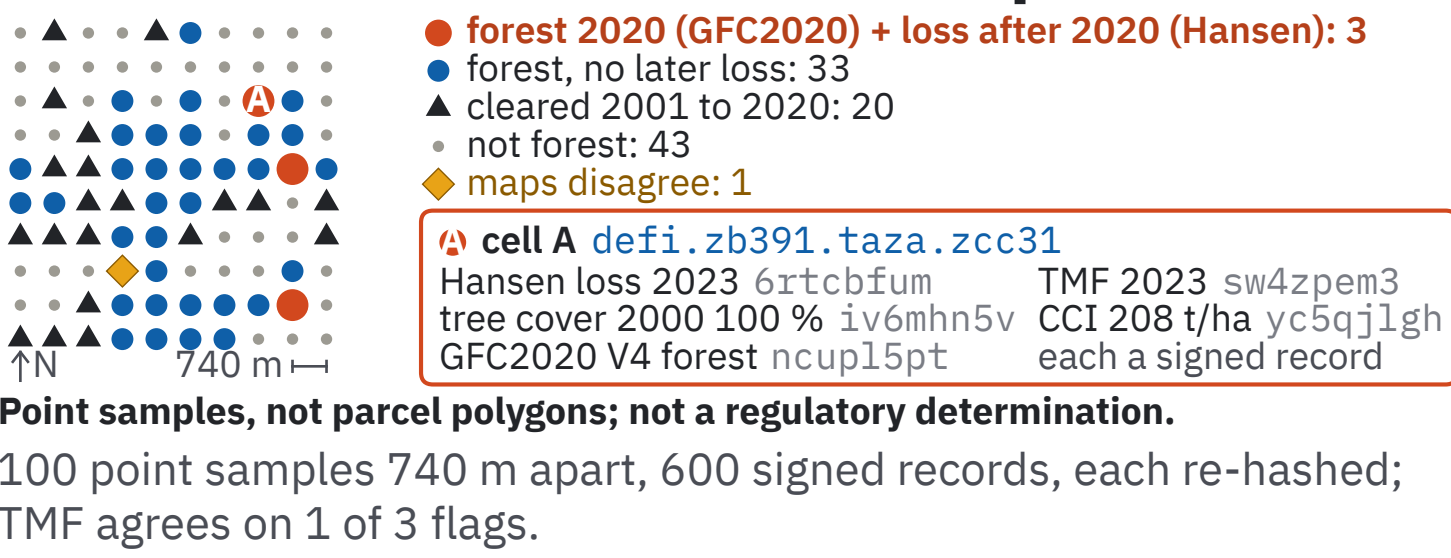
8 What did the agent know on 15 Jun?

### Memory keeps what was cited



9 Can a screen be re-checked input by input?

### An auditor re-runs each input



10 Where are the foundation-model embeddings?

### Vectors are records too

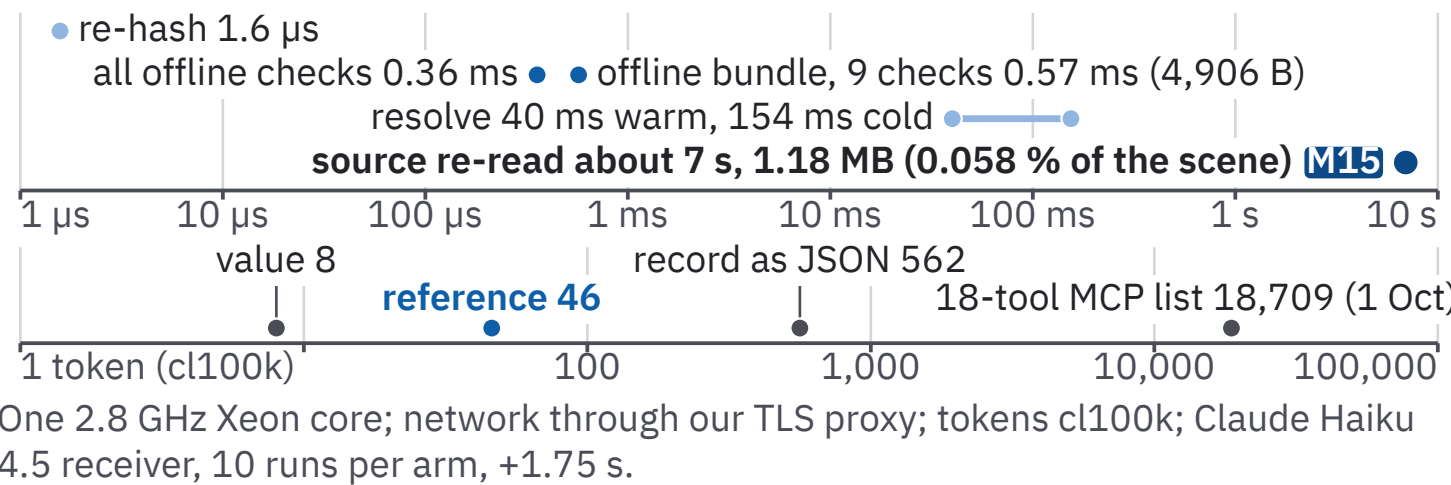
Derived representations are addressable too: this cell's Prithvi-EO-2.0 vector names its checkpoint digest inside the hashed record; the TESSERA vector, only a path and year. Both still resolve; emem.dev lists its encoders as retired.



11 What does checking cost?

### The source re-read costs most

A record check takes 0.36 ms of CPU; a pixel re-read moves 1.18 MB in about 7 s, and only it refuses the wrong pixel. A reference costs 46 tokens, the value 8; a model receiver paid 2.1 times the input tokens for the same decisions.



## 13 EMEM relies on these layers

A STAC item identifies a file; EMEM identifies the observation an agent cited, for the next agent to re-check. GeoGuard judges the claim; EMEM fixes the evidence it cites.

|                |                         |                            |
|----------------|-------------------------|----------------------------|
| JUDGE          | GeoGuard                | a claim in text            |
| CARRY          | MCP, A2A                | a tool call, an agent card |
| RETRIEVE       | RAG                     | a text chunk               |
| SIGN FILES     | C2PA, CDSE Traceability | a file                     |
| RECORD LINEAGE | W3C PROV                | entity, activity, agent    |
| RUN            | openEO                  | a process graph            |
| FIND           | STAC                    | an asset (file)            |

A signed observation reference left the receiver acting on none of the 16 corruptions in our suite; a source re-read exposes a wrong pixel emem signed in production; entity, sensor and decision stay beyond its checks.  
READ THE METHODS vortex-ai.github.io/esa-poster  
Definitions, denominators, models, and what our tools generated.

